LANGUAGE TRANSLATORS

ASSIGNMENT -1

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System S/W and Machine Architecture

System S/W is a machine dependent instruction related to the architecture of the m/c.

S/c (Simplified Instruction computer)

S/C has been carefully designed to include the H/w features. mostly . found on many computes
s/c has 2 versions s/c has 2 versions

1. Standard Models (It has inst of same length)
2. XE version (Extra equipment) or (Extra Expensive)
(It has diff inst length)

S/C M/c architecture

The S/C M/c architecture depends on the following features.

- Memory
- Registers
- Data formats
- Instruction formats
- Addressing modes
- Instruction set
- I/P and O/P

Memory : Supported by SIC

- Memory consists of 8 bit bytes
- Any 3 consecutive bytes from a word (24 bits)
- Total of 32768 (2^{18}) bytes in the computer memory.
- All address are byte addressed

12345678	9. . . . 16	17. 24

← 1 word →

Registers

There are 5 Registers each register has 24 bits.

Mnemonic	Number	Use
A	0	Accumulator : used for arithmetic operations
X	1	Index register : used for addressing
L	2	Linkage register : contains the return address
PC	8	Program counter : contains the address of the next instruction to be executed
SW	9	Status word, including Condition code such as <, ≤, >, ≥, == .

Data formats

- Integers are stored as 24 bit binary number
- 2's complement representation for negative values
- Characters are stored using 8 bit ASCII codes
- No floating point hardware on the standard version of SIC.

Instruction Formats

- All machine instructions on the standard version of SIC have the 24 bit format as shown

Opcode (8)	X	Address (15)

Addressing Mode

- There are two addressing modes available.

Mode	Indication	Target address calculation
Direct	x=0	TA = address
Indexed	x=1	TA = address + (X)

Instruction set

- Load and store registers - LDA, LDX, STA, STX, etc
- Integer arithmetic operations - ADD, SUB, MUL, DIV
- COMP - comparision instruction

- Conditional jump instructions - JLT, JEQ, JGT
- Subroutine linkage - JSUB, RSUB
- I/O
  - Test Device instruction (TD)
  - Read Data (RD)
  - Write Data (WD)

SIC / XE Machine Architecture

Memory

- Memory consists of 8 bit bytes
- Any 3 consecutive bytes form a word (24 bits)
- Total of 1Mb (2^{20}) bytes in the computer memory.

Registers

There are nine registers, each of 24 bits in length.

Mnemonics	Number	Use.
A	0	Accumulator: used for arithmetic operations.
X	1	Index register; used for addressing
L	2	Linkage register: contains the return address whenever control transferred to subroutine.

B	3	Base register : used for addressing
S	4	General working register - no special use
T	5	General working register - no special use
F	6	Floating point accumulator (48 bit
PC	8	Program counter: contains the address of the next instruction. to be executed.
SW	9	Status word, including, condition code <, ≤, >, ≥, ==

Data Formats

- Integers are stored as 24 bit binary number
- 2's complement representation for negative values
- Characters are stored using 8 bit ASCII codes.
- Support 48 bit floating point numbers.

1	11	36
S	exponent	fraction

- There is a 48 bit floating point data type,

$F * 2^{(e-1024)}$

Instruction formats

Format 1 (1 byte) -

8
opcode

Format 1 (2 byte) -

8	4	4
op	r1	r2

Format 3 (3 byte)

6	1	1	1	1	1	1	12
op	n	i	x	b	p	e	Displacement

Format 4 (4 byte).

6	1	1	1	1	1	1	20
op	n	i	x	b	p	e	Address

Addressing modes

- Base relative (n=1, i=1, b=1, p=0)
- Program counter relative (n=1, i=1, b=0, p=1)
- Direct (n=1, i=1, b=0, p=0)
- Immediate (n=0, i=1, x=0)
- Indirect (n=1, i=0, x=0)
- Indexing (both n & i = 0 or 1, x=1)
- Extended (e=1 for format 4, e=0 for format 3)

Instruction set

- Load and store registers LDA, LDX, STA, STX, LDB, STB etc.

- Integer arithmetic operations - ADD, SUB, MUL, DIV

- Floating point arithmetic operations: ADDF, SUBF, MULF, DIVF.

- COMP - Comparision instruction.

- Conditional jump instructions - JLT, JEQ, JGT

- Subroutine linkage - JSUB, RSUB

- Register move instruction RMO

- Register to register arithmetic operations ADDR, SUBR, MULR, DIVR.

- I/O
  - Test Device instruction (TD)
  - Read Data (RD)
  - Write Data (WD)